

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Zinc chloride, 2M solution in 2-methyltetrahydrofuran

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>428560000; 428561000; 428568000</b>                                                               |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

|                   |            |
|-------------------|------------|
| Flammable liquids | Category 1 |
|-------------------|------------|

#### Health hazards

|                                                    |              |
|----------------------------------------------------|--------------|
| Acute Oral Toxicity                                | Category 4   |
| Skin Corrosion/Irritation                          | Category 1 B |
| Serious Eye Damage/Eye Irritation                  | Category 1   |
| Specific target organ toxicity - (single exposure) | Category 3   |

#### Environmental hazards

|                          |            |
|--------------------------|------------|
| Acute aquatic toxicity   | Category 1 |
| Chronic aquatic toxicity | Category 1 |

### Label Elements



Flame



Exclamation Mark



Corrosion



Environment

**Signal Word**

**Danger**

**Hazard Statements**

H224 - Extremely flammable liquid and vapor  
H302 - Harmful if swallowed  
H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation  
H410 - Very toxic to aquatic life with long lasting effects  
AUH019 - May form explosive peroxides

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P372 - Explosion risk  
P374 - Fight fire with normal precautions from a reasonable distance  
P380 - Evacuate area  
P403 + P235 - Store in a well-ventilated place. Keep cool  
P404 - Store in a closed container  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component             | CAS No    | Weight % |
|-----------------------|-----------|----------|
| Methyltetrahydrofuran | 96-47-9   | 75       |
| Zinc chloride         | 7646-85-7 | 25       |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required. If not breathing, give artificial respiration.                                                                                                                |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                             |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                              |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Difficulty in breathing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                                                                                                                                                                                                                                                                        |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Hazardous Decomposition Products

Hydrogen chloride gas.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. May form explosive peroxides. Do not allow run-off from fire-fighting to enter drains or water courses.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

### Methods for Containment and Clean Up

#### Clean-up methods - small spillage

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use

spark-proof tools and explosion-proof equipment.

#### Clean-up methods - large spillage

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

#### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

#### Precautions for Safe Handling

Use only under a chemical fume hood. Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance. Take precautionary measures against static discharges. If peroxide formation is suspected, do not open or move container. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded.

#### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Flammables area. Corrosives area. Keep away from heat, sparks and flame. Store under an inert atmosphere. Containers should be dated when opened and tested periodically for the presence of peroxides. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

#### Exposure limits

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component     | Australia                                             | New Zealand WEL                                       | ACGIH TLV                                             | The United Kingdom                                                | Germany                                                                                                                                                                                          |
|---------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc chloride | STEL: 2 mg/m <sup>3</sup><br>TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | STEL: 2 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr | TWA: 0.1 mg/m <sup>3</sup> (8 Stunden). MAK<br>TWA: 2 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 2 mg/m <sup>3</sup><br>Höhepunkt: 0.4 mg/m <sup>3</sup><br>Höhepunkt: 4 mg/m <sup>3</sup> |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

#### Exposure Controls

##### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or

equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-------------------|-----------------|-----------------|-----------------------|
| Natural rubber | See manufacturers | -               | AS/NZS 2161     | (minimum requirement) |
| Nitrile rubber | recommendations   |                 |                 |                       |
| Neoprene       |                   |                 |                 |                       |
| PVC            |                   |                 |                 |                       |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:**

Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-**

Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

**Information on basic physical and chemical properties**

|                              |                          |                                   |
|------------------------------|--------------------------|-----------------------------------|
| Appearance                   | Colorless                |                                   |
| Physical State               | Liquid                   |                                   |
| Odor                         | No information available |                                   |
| Odor Threshold               | No data available        |                                   |
| pH                           | No information available |                                   |
| Melting Point/Range          | No data available        |                                   |
| Softening Point              | No data available        |                                   |
| Boiling Point/Range          | No information available |                                   |
| Flash Point                  | -11 °C / 12.2 °F         | Method - No information available |
| Evaporation Rate             | No data available        |                                   |
| Flammability (solid,gas)     | Not applicable           | Liquid                            |
| Explosion Limits             | No data available        |                                   |
| Vapor Pressure               | No data available        |                                   |
| Vapor Density                | No data available        | (Air = 1.0)                       |
| Specific Gravity / Density   | 1.07                     |                                   |
| Bulk Density                 | Not applicable           | Liquid                            |
| Water Solubility             | No information available |                                   |
| Solubility in other solvents | No information available |                                   |

**Partition Coefficient (n-octanol/water)**

**Autoignition Temperature** No data available

**Decomposition Temperature** No data available

**Viscosity** No data available

**Explosive Properties**

**Oxidizing Properties** No information available

Vapors may form explosive mixtures with air

**Other information**

**Molecular Formula** Cl<sub>2</sub> Zn

**Molecular Weight** 136.29

## Section 10 - Stability and Reactivity

**Reactivity** Yes

**Stability** May form explosive peroxides.

**Conditions to Avoid** Incompatible products, Excess heat, Keep away from open flames, hot surfaces and sources of ignition.

**Incompatible Materials** Strong oxidizing agents.

**Hazardous Decomposition Products** Hydrogen chloride gas.

**Hazardous Polymerization** Hazardous polymerization does not occur.

## Section 11 - Toxicological Information

**Information on Toxicological Effects**

**Product Information**

**(a) acute toxicity;**

**Oral** Category 4

**Dermal** Based on available data, the classification criteria are not met

**Inhalation** Based on available data, the classification criteria are not met

**Toxicology data for the components**

| Component             | LD50 Oral              | LD50 Dermal           | LC50 Inhalation                               |
|-----------------------|------------------------|-----------------------|-----------------------------------------------|
| Methyltetrahydrofuran | 300-2000 mg/kg ( Rat ) | 4500 mg/kg ( Rabbit ) | 6000 ppm ( Rat ) 4 h                          |
| Zinc chloride         | 350 mg/kg (Rat)        |                       | LC50 <= 1975 mg/m <sup>3</sup> ( Rat ) 10 min |

**(b) skin corrosion/irritation;** Category 1 B

**(c) serious eye damage/irritation;** Category 1

**(d) respiratory or skin sensitization;**

**Respiratory** No data available

**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

|                                            |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                            | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                                                                                                                                       |
| (g) reproductive toxicity;                 | No data available                                                                                                                                                                                                                                                                                                                                                               |
| (h) STOT-single exposure;                  | Category 3                                                                                                                                                                                                                                                                                                                                                                      |
| Results / Target organs                    | Respiratory system                                                                                                                                                                                                                                                                                                                                                              |
| (i) STOT-repeated exposure;                | No data available                                                                                                                                                                                                                                                                                                                                                               |
| Target Organs                              | No information available.                                                                                                                                                                                                                                                                                                                                                       |
| (j) aspiration hazard;                     | No data available                                                                                                                                                                                                                                                                                                                                                               |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated.                                                                                                                                                                                                                                                                                                                  |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting |

## Section 12 - Ecological Information

**Ecotoxicity effects** The product contains following substances which are hazardous for the environment. Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment.

| Component             | Freshwater Fish                                                  | Water Flea                                              | Freshwater Algae                                   | Microtox |
|-----------------------|------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|----------|
| Methyltetrahydrofuran | LC50 (96h) > 100 mg/l<br>Onchorhynchus mykiss<br>(Rainbow trout) | Chronic NOEC >= 120<br>mg/l (21 days, Daphnia<br>magna) | NOEC >= 104 mg/l<br>(72h)<br>EC50 > 104 mg/l (72h) |          |
| Zinc chloride         | LC50: 0.4-2.2 mg/L/96h<br>(Cyprinus carpio)                      | EC50: 0.2 mg/L/48h                                      | EC50: 0.027-0.105<br>mg/L/72h                      |          |

**Persistence and Degradability** No information available

| Component                             | Degradability |
|---------------------------------------|---------------|
| Methyltetrahydrofuran<br>96-47-9 (75) | (2%) 28 days  |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** No information available

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Zinc chloride |         | 16000 dimensionless           |

**Mobility** No information available.

**Endocrine Disruptor Information**

**Persistent Organic Pollutant**

This product does not contain any known or suspected substance

**Ozone Depletion Potential**

This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

|                          |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Other Information</b> | Chemical wastes should be disposed through a licensed commercial waste collection service. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. Do not let this chemical enter the environment. |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                      |
|--------------------------------|--------------------------------------|
| <b>UN-No</b>                   | UN2924                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.  |
| <b>Technical Shipping Name</b> | Methyltetrahydrofuran, Zinc chloride |
| <b>Hazard Class</b>            | 3                                    |
| <b>Subsidiary Hazard Class</b> | 8                                    |
| <b>Packing Group</b>           | II                                   |

| Component                         | IMDG Marine Pollutant                    |
|-----------------------------------|------------------------------------------|
| Zinc chloride<br>7646-85-7 ( 25 ) | IMDG regulated marine pollutant (UN1840) |

### ADG

|                                |                                      |
|--------------------------------|--------------------------------------|
| <b>UN-No</b>                   | UN2924                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.  |
| <b>Technical Shipping Name</b> | Methyltetrahydrofuran, Zinc chloride |
| <b>Hazard Class</b>            | 3                                    |
| <b>Subsidiary Hazard Class</b> | 8                                    |
| <b>Packing Group</b>           | II                                   |

| Component                         | Hazchem Code |
|-----------------------------------|--------------|
| Zinc chloride<br>7646-85-7 ( 25 ) | 2X           |

### IATA

|                                |                                      |
|--------------------------------|--------------------------------------|
| <b>UN-No</b>                   | UN2924                               |
| <b>Proper Shipping Name</b>    | Flammable liquid, corrosive, n.o.s.  |
| <b>Technical Shipping Name</b> | Methyltetrahydrofuran, Zinc chloride |
| <b>Hazard Class</b>            | 3                                    |
| <b>Subsidiary Hazard Class</b> | 8                                    |
| <b>Packing Group</b>           | II                                   |

|                              |                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Environmental hazards</b> | Dangerous for the environment<br>Product is a marine pollutant according to the criteria set by IMDG/IMO |
|------------------------------|----------------------------------------------------------------------------------------------------------|

|                            |                                 |
|----------------------------|---------------------------------|
| <b>Special Precautions</b> | No special precautions required |
|----------------------------|---------------------------------|

|                               |            |
|-------------------------------|------------|
| <b>Additional information</b> | None known |
|-------------------------------|------------|

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                             |           |
|-----------------------------|-----------|
| <b>National Regulations</b> | Australia |
|-----------------------------|-----------|

See section 8 for national exposure control parameters.

Standard for the Uniform Scheduling of Medicines and Poisons



Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                 | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zinc chloride - 7646-85-7 | Schedule 2 listed<br>Schedule 4 listed - for human internal use except in preparations with a recommended daily dose of <=25 mg of Zinc, or in preparations with a recommended daily dose of between 25-50 mg of Zinc when compliant with the requirements of the Required Advisory Statements for Medicine Labels<br>Schedule 6 listed - except when included in Schedule 2, or in preparations containing <=5% of Zinc chloride |

**Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                       | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|---------------------------------|-------------------------------------------------------------|------------------------|
| Methyltetrahydrofuran - 96-47-9 | Present                                                     | -                      |
| Zinc chloride - 7646-85-7       | Present                                                     | -                      |

**Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

**Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

**National pollutant inventory** Not applicable

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component             | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-----------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Methyltetrahydrofuran | X    | X     | 202-507-4 | -      | X    | X   | -    | X     | -    | X    | X     | KE-33479 |
| Zinc chloride         | X    | X     | 231-592-0 | -      | X    | X   | -    | X     | X    | X    | X     | KE-35535 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**MARPOL** - International Convention for the Prevention of Pollution from Ships

| Component                 | IMDG Marine Pollutant                    |
|---------------------------|------------------------------------------|
| Zinc chloride - 7646-85-7 | IMDG regulated marine pollutant (UN1840) |

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Take note that wastes may be subject to export, import, or transit controls pursuant to the Basel convention and/or local regulations implementing the Basel convention.

| Component                 | Basel Convention (Hazardous Waste) | Australian Hazardous Waste Act - Categories of Wastes to Be Controlled |
|---------------------------|------------------------------------|------------------------------------------------------------------------|
| Zinc chloride - 7646-85-7 | Annex I - Y23                      | Y23                                                                    |

| Component             | CAS No    | OECD HPV       | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------------|-----------|----------------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyltetrahydrofuran | 96-47-9   | Not applicable | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |
| Zinc chloride         | 7646-85-7 | Listed         | Not applicable                             | Not applicable                                                                            | Not applicable                                                                           |

#### Authorisation/Restrictions according to EU REACH

| Component     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Zinc chloride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Section 16 - Other Information

### Legend

**AICS** - Australian Inventory of Chemical Substances  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**NZS 5433:2012** - Transport of Dangerous Goods on Land

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**NZIoC** - New Zealand Inventory of Chemicals

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

Predicted No Effect Concentration (PNEC)

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**ADG** Australian Code for the Transport of Dangerous Goods by Road and Rail

**OECD** - Organisation for Economic Co-operation and Development

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

#### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

#### Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

#### Training Advice

Chemical incident response training.

Revision Date 17-Nov-2022  
Revision Summary Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

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**End of Safety Data Sheet**